TURNING 700 ON:

1. Push Switch (located on back)
   Push → ON
   Push → OFF

2. Depress PRIME, Set RUN Mode.

MODES OF OPERATION

RUN | LEARN | LEARN & PRINT | LIST PROG.



RUN: For normal operation & program execution.

LEARN: For entering a program into core from the key-board.

LEARN / PRINT: For entering a pro-gram into core from the key-board while simultaneously printing each step on typewriter.

LIST PROGRAM: For listing current program in core on typewriter. Depress GO to list step number and program code.

700 PROGRAM CODES

CODE	KEY
0400	+ DIRECT
0401	– DIRECT
0402	× DIRECT
0403	÷ DIRECT
0404	STORE DIRECT
0405	RECALL DIRECT
0406	↔ DIRECT
0407	SEARCH
0408	MARK
0409	GROUP 1
0410	GROUP 2
0411	WRITE
0412	WRITE ALPHA
0413	END ALPHA
0414	STORE Y
0415	RECALL Y

CODE	KEY
0500	+ INDIR
0501	– INDIR
0502	× INDIR
0503	÷ INDIR
0504	STORE INDIR
0505	RECALL INDIR
0506	↔ INDIR
0507	SKIP IF Y ≥ X
0508	SKIP IF Y < X
0509	SKIP IF Y = X
0510	SKIP IF ERROR
0511	RETURN
0512	END PROG
0513	LOAD PROG
0514	GO
0515	STOP

CODE	KEY
0600	+
0601	–
0602	×
0603	÷
0604	↑
0605	↓
0606	↕
0607	|X|
0608	INTEGER X
0609	π
0610	$Log_{10}X$
0611	$Log_{e}X$
0612	$\sqrt{X}$
0613	10^{X}
0614	e^{X}
0615	1/X

CODE	KEY
0700	0
0701	1
0702	2
0703	3
0704	4
0705	5
0706	6
0707	7
0708	8
0709	9
0710	SET EXP
0711	CHANGE SIGN
0712	DECIMAL POINT
0713	X^2
0714	RECALL RESIDUE
0715	CLEAR X

TAPE DRIVE OPERATION

RELEASE | FORWARD | TAPE READY | REWIND



RELEASE: Releases tape cassette for removal.

FORWARD: Moves tape forward when depressed.

TAPE READY: Places head of tape reader in contact with tape.

REWIND: Rewinds tape when depressed.

LOADING
RECORDING
PROGRAMMING
NON-PROG.
ERROR LIGHTS

TO RECORD PROGRAM
(Core to Tape)

1. Insert tape cassette. REWIND, (if necessary).
2. Depress TAPE READY.
3. SET PC to first step of program (or PRIME).
4. Depress RECORD PROG.

TO "LEARN" PROGRAM
(Keyboard Entry)

1. Depress LEARN mode.
2. SET PC to desired step (OR PRIME).
3. Depress keys in program.
4. Depress END PROG as final step.

TO LOAD PROGRAM
(Tape to Core)

1. Depress RUN mode.
2. Insert tape cassette. REWIND (if necessary).
3. Depress TAPE READY.
4. SET PC to desired step number (or PRIME).
5. Depress LOAD PROG.

TO RECORD DATA ON TAPE
(Core to Tape)

1. Insert tape cassette REWIND (if necessary).
2. Depress TAPE READY.
3. SET P.C. To step which starts desired register. (See table of core in manual.) END PROG necessary at last step in register to be recorded.
4. Depress RECORD PROG.

PROGRAMMING KEYS

LOAD PROG Transfer Program from tape to core. 0513	**SKIP IF ERROR** Skips 2 steps if condition met 0510	**MARK** Mark followed by any code sets the flag at that step. 0408
END PROG Defines Prog Block. 0512	**SKIP IF Y ≥ X** Skips 2 steps if condition met 0507	**RETURN** Branches back from a sub-routine. 0511
STOP Stops program Execution. 0515	**SKIP IF Y = X** Skips 2 steps if condition met 0509	**GROUP 1** Addresses Peripheral equipment. 0409
GO Continues execution of a program after a stop. May be used as "no operation" step. 0514	**SKIP IF Y < X** Skips 2 steps if condition met 0508	**GROUP 2** Addresses Peripheral equipment. 0410
	SEARCH Search followed by any code branches unconditionally to corresponding mark. 0407	

NON-PROGRAMMABLE KEYS

Clear X & Y sets program counter to 000 and turns error light off.	**PRIME**
Sums all program codes from step 000 to step preceding END PROG.	**VERIFY PROG.**
Sets program counter with next 3 keystrokes.	**SET PC**
Transfers program from core to tape.	**RECORD PROG.**
In RUN-mode interrupts program execution and allow single step execution of program. In LEARN mode displays step number and code of each command.	**STEP**

720 models use 4 digit PC.

ERROR LIGHTS
(Sign in display flashing)



PROGRAM ERROR

Turned on when illegal programming operation is performed or calculation results exceeds 10^{99}. Can be tested using SKIP IF ERROR decision.



MACHINE ERROR

Turned on when program has not been transferred properly from tape cassette. REPEAT TAPE OPERATION.

WRITE ALPHA Sets unit for alpha printout 0412	**END ALPHA** Ends alpha printout 0413
WRITE Used for numeric printout & format control 0411	$\frac{1}{X}$ 0615
INTEGER X Truncates X leaving integer part in X 0608	**|X|** 0607
10^X 0613	$LOG_{10}X$ 0610
e^X 0614	LOG_eX 0611

RECALL RESIDUE 0714	Y into X Y unchanged 0605
π Displays π in X 0609	
↔ Interchanges Y with X 0606	X into Y X unchanged 0604

RECALL RESIDUE

Used for double precision arithmetic to obtain another 12 digits of accuracy when adding, subtracting, multiplying. After division, remainder in X.

EXAMPLE: ADD

5 0 2 4 8 7 3 0 5 8 . 2 8

+

6 . 8 5 2 0 9 8 7

+

Read

.502487306513 +10 in Y

RECALL RESIDUE

Read remaining digits

.209870000000 –02 in X

CHANGE SIGN 0711	$\sqrt{X}$ 0612	X^2 0713	**CLEAR X** 0715
÷ Y ÷ X → Y X unchanged 0603	**7** 0707	**8** 0708	**9** 0709
× Y × X → Y X unchanged 0602	**4** 0704	**5** 0705	**6** 0706
– Y – X → Y X unchanged 0601	**1** 0701	**2** 0702	**3** 0703
+ Y + X → Y X unchanged 0600	**0** 0700	**▪** 0712	**SET EXP** Indicates digits to follow affects exponent of X. 0710

NUMERIC ENTRY & BASIC ARITHMETIC OPERATIONS USING X & Y

1. $Log_{10}\ 256 =$

Index 2 5 6

$Log_{10}X$

Read 2.40823996532 in X

2. $e^{5.23} =$

5 . 2 3

e^X

Read 186.792803518 in X

3. $71 + 22 + Log_e\ 13 =$

Index 7 1 + 2 2 +

1 3 Log_eX +

Read 95.5649493574 in Y

4. $\sqrt{\frac{55 \times .876 \times 10^{-5}}{63}}$

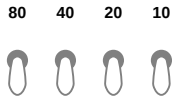
Index 5 5 ↑ 8 7 6 CHS SET EXP 5

× 6 3 ÷ ↓ $\sqrt{X}$

Read .276543288611 –02 in X

Typos/errors in originals, not corrected here.

TOGGLE SWITCHES AND SPECIAL OPERATION KEYS



00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15

The Toggle Switches and Special Operation Keys are used for three basic purposes:

1. To address subroutines (such as TRIG and STAT functions) with 1 keystroke.
2. To address 120 data registers.
3. To generate any 700 program code.
4. To format output of typewriter.

EXAMPLE: Find SIN 30° (if TRIG PACK loaded)

Index 3 0 **02**

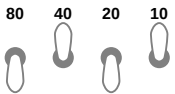
Read .500000000000 in X

NOTE: To protect the loaded functions by pressing VERIFY PROG before learning additional program steps.

122 DATA REGISTERS (000 to 121) Which can be used to add, subtract, multiply, and divide. A register is designated in 1 of 2 ways.

1. Indirect addressing by integer number in Y.
2. Direct Addressing by sum of toggle switches and special operation keys.

STORE DIRECT



Depress STORE DIRECT **06**

Register 56 is addressed.

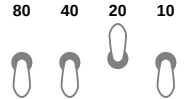
NOTE: After arithmetic operations (DIRECT & INDIRECT) answer is placed in internal register.

- 720 models have 250 registers and add "+100" variants of DIRECT commands (code + 0800).
- B and C models may use the last 2 registers during certain commands, leaving 120/248.

RECALL INDIRECT Recalls to X, REG. indicated in Y 0505	**INDIRECT** Exchanges X with REG. indicated in Y 0506	**DIRECT** Exchanges X with REG. designated by next key 0406	**RECALL DIRECT** Recalls to X, REG. designated by next key 0405
STORE INDIRECT Store X into REG. indicated in Y 0504	÷ **INDIRECT** Divides X into REG. indicated in Y 0503	÷ **DIRECT** Divides X into REG. designated next key 0403	**STORE DIRECT** Stores X, into REG. designated by next key 0404
	× **INDIRECT** Miltiplies X by REG. indicated in Y 0502	× **DIRECT** Multiplies X by REG. designated by next key 0402	
	– **INDIRECT** Subtracts X from REG. indicated in Y 0501	– **DIRECT** Subtracts X from REG. designated by next key 0401	
	+ **INDIRECT** Adds X to REG. indicated in Y 0500	+ **DIRECT** Adds X to REG. designated by next key 0400	

EXAMPLE:
Perform 12 × 2
in REG. 21 6

DIRECT ADDRESSING
(2 step command)
Toggle Switch Setting



1 2 STORE DIRECT 01
2 × DIRECT 01
6 ÷ DIRECT 01
RECALL DIRECT 01

INDIRECT ADDRESSING
(1 step command)
2 1 ↑
1 2 ST INDIR
2 × INDIR
6 ÷ INDIR
RE INDIR

Read 4 in X